PCB Redesign via Reverse Engineering

Legacy Board Recovery & Improvement Project

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1. Project Overview

This project involved the **reconstruction and redesign of an existing PCB** for which **complete schematics and documentation were unavailable**.
The objective was to **recover the circuit intent**, rebuild missing design information, and produce a **functionally equivalent and improved PCB layout** suitable for continued use and future modification.

The work required **systematic reverse engineering**, analysis of existing routing and component placement, and validation through redesign.

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2. Problem Statement

The original PCB presented several constraints:

- Missing or incomplete schematic documentation
- Limited design rationale available for routing and placement decisions
- Legacy layout not optimized for maintainability or modification
- Need to preserve functional behavior while enabling future changes

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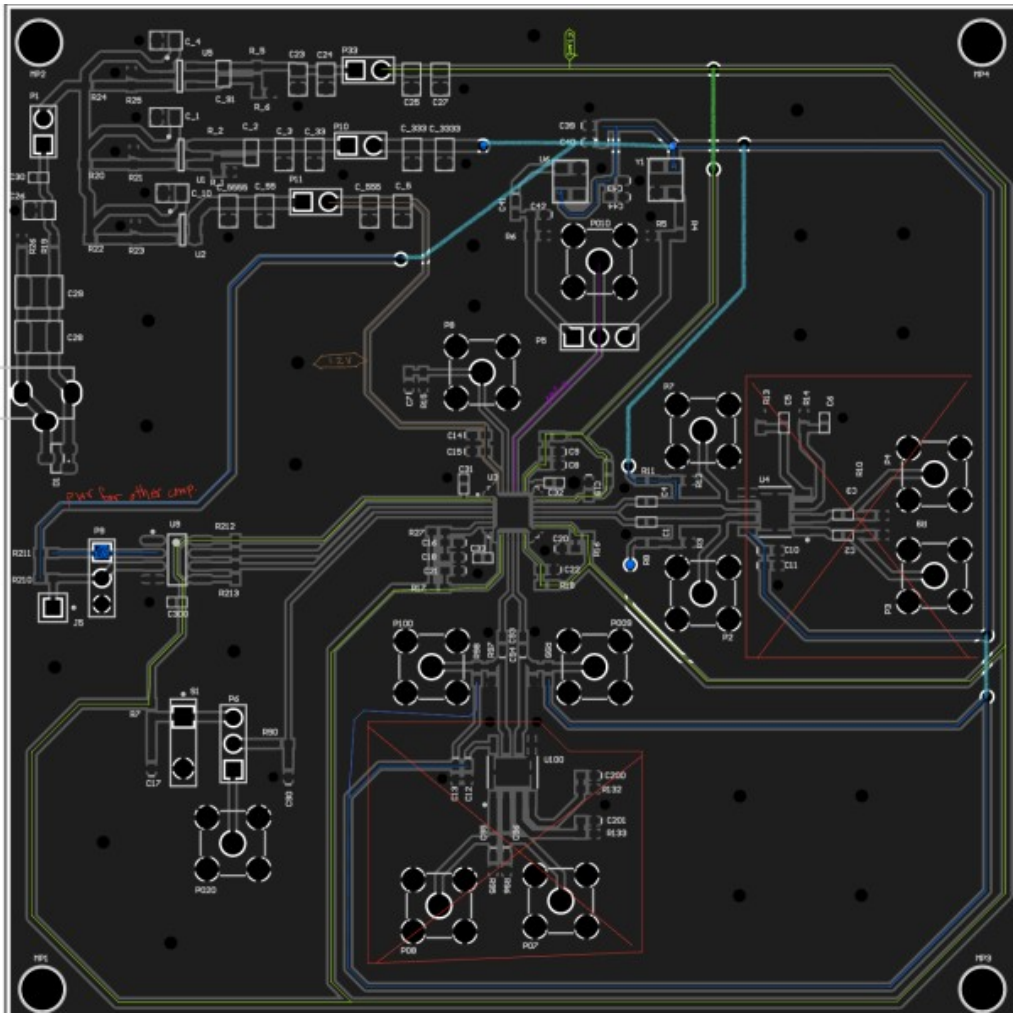
3. Reverse Engineering Methodology

The redesign process followed a structured engineering approach:

1. **Physical and visual inspection** of the existing PCB
2. Identification of functional blocks based on component grouping and routing patterns
3. Net tracing and reconstruction of signal connectivity
4. Inference of power, analog, and digital domains
5. Re-creation of schematic intent sufficient to support redesign

This process focused on understanding **why** the original design behaved as it did, not merely replicating geometry.



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4. Redesign Objectives

Based on the recovered design intent, the redesign targeted:

- Functional equivalence with the original board
- Improved routing clarity and structure
- Better separation of functional domains
- Increased readability and maintainability
- Easier support for future revisions or debugging

The goal was not blind duplication, but **engineering improvement under constraint**.

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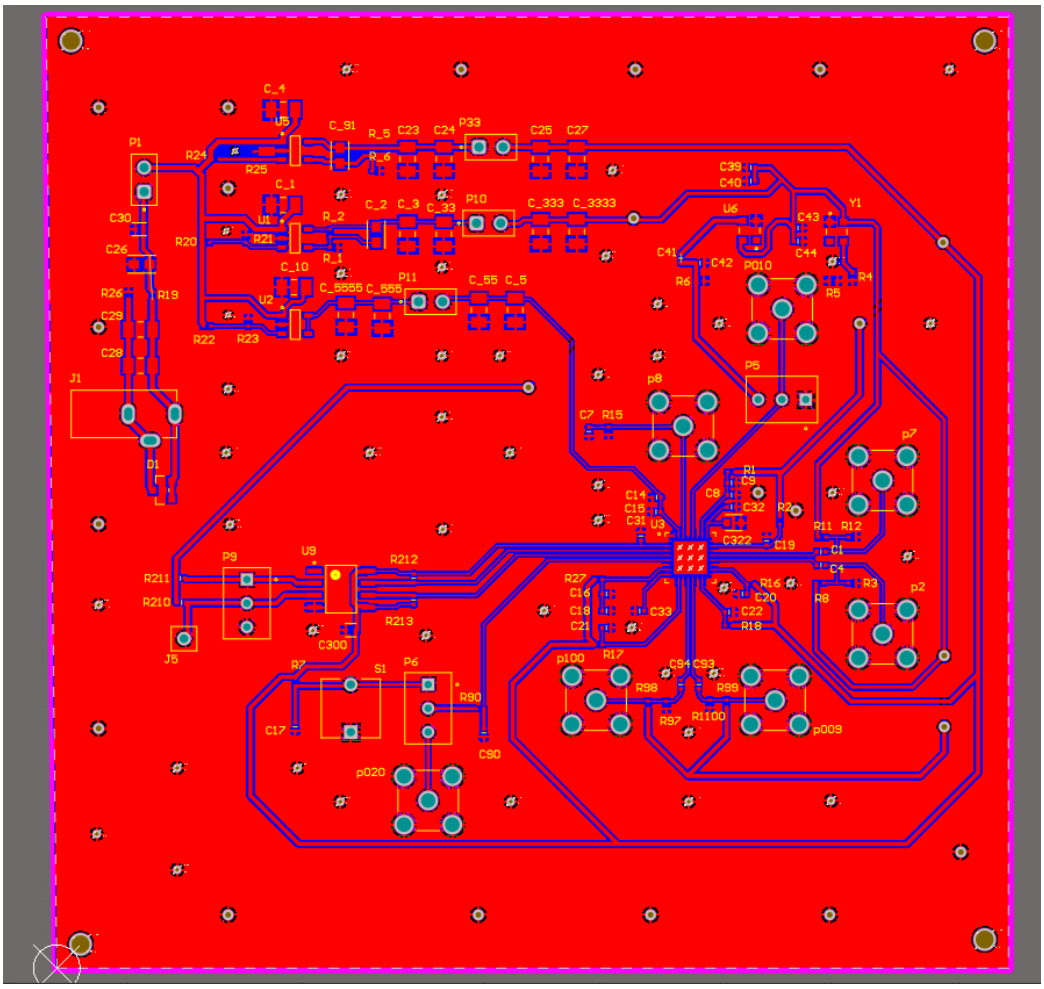
5. New PCB Design & Layout

A new PCB layout was developed:

- Improved grounding and power distribution clarity
- Simplified routing paths where possible without altering behavior

Design decisions were made conservatively to preserve expected electrical behavior.



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6. Engineering Challenges

Key challenges addressed during the project included:

- Interpreting undocumented routing choices
- Avoiding unintended behavioral changes
- Balancing fidelity to the original design with layout improvements
- Maintaining electrical assumptions without full specifications

These challenges required careful reasoning, iterative verification, and disciplined design judgement.

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7. Verification Intent

While the project focused on redesign rather than new functionality, the PCB was prepared to support:

- Functional equivalence testing
- Debugging and probing
- Incremental modification without full rework

The redesigned board provides a clearer platform for future engineering work.

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8. Confidentiality & Ethics Statement

This project involved **legitimate reverse engineering for recovery, maintenance, or redesign purposes**.
No proprietary documentation, firmware, or confidential design files were accessed.

The redesigned PCB reflects **independent engineering work** authorized by the original author of the board, derived from analysis and reconstruction, not direct copying.

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9. Conclusion

This project demonstrates capability in:

- Reverse engineering undocumented PCB designs
- Recovering schematic intent from physical layouts
- Redesigning boards for clarity and maintainability
- Applying disciplined engineering judgment under incomplete information

Such skills are critical in legacy system support, hardware maintenance, and real-world product development environments.